KAIZEN MATHS

Lesson starter

Factorising Quadratic Expressions

Coefficient of x squared

Name

S1

Factorise each expression.

Factorise: $4x^2 - 33x + 8$

S2

Factorise each expression.

Factorise: $15x^2 + 58x + 48$

S3

Factorise each expression.

Factorise: $10x^2 + 33x - 7$

S4

Factorise each expression.

Factorise: $5x^2 - 13x - 6$

Show your working

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ALG-CFE2-L2 · S

Show your working

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Exit ticket

Factorising Quadratic Expressions

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Factorise each expression.

Factorise: $15x^2 - 46x + 35$

E2

Factorise each expression.

Factorise: $30x^2 - 37x - 7$

E3

Factorise each expression.

Factorise: $3x^2 - 19x + 28$

ALG-CFE2-L2 · E

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ALG-CFE2-L2 · E

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Teacher answers & working

Factorising Quadratic Expressions · Coefficient of x squared

ALG-CFE2-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Factorise each expression.

Factorise: $4x^2 - 33x + 8$

Answer:

$(4x - 1)(x - 8)$

Working:

Use the product-sum test before writing the brackets.

$a = 4, \quad b = -33, \quad c = 8$

$a \times c = 4 \times 8 = 32$

$-32 \times -1 = 32 \quad \text{and} \quad -32 + -1 = -33$

Split the middle term using those two numbers.

$4x^2 - 33x + 8$

$= 4x^2 - 32x - x + 8$

Group the expression in pairs, then factor each group.

$= 4x(x - 8) - (x - 8)$

$= (4x - 1)(x - 8)$

S2

Factorise each expression.

Factorise: $15x^2 + 58x + 48$

Answer:

$(3x + 8)(5x + 6)$

Working:

Use the product-sum test before writing the brackets.

$a = 15, \quad b = 58, \quad c = 48$

$a \times c = 15 \times 48 = 720$

$18 \times 40 = 720 \quad \text{and} \quad 18 + 40 = 58$

Split the middle term using those two numbers.

$15x^2 + 58x + 48$

$= 15x^2 + 18x + 40x + 48$

Group the expression in pairs, then factor each group.

$= 3x(5x + 6) + 8(5x + 6)$

$= (3x + 8)(5x + 6)$

Teacher answers & working

Factorising Quadratic Expressions · Coefficient of x squared

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S3

Factorise each expression.

Factorise: $10x^2 + 33x - 7$

Answer:

$(5x - 1)(2x + 7)$

Working:

Use the product-sum test before writing the brackets.

$a = 10, \quad b = 33, \quad c = -7$

$a \times c = 10 \times -7 = -70$

$35 \times -2 = -70 \quad \text{and} \quad 35 + -2 = 33$

Split the middle term using those two numbers.

$10x^2 + 33x - 7$

$= 10x^2 + 35x - 2x - 7$

Group the expression in pairs, then factor each group.

$= 5x(2x + 7) - (2x + 7)$

$= (5x - 1)(2x + 7)$

S4

Factorise each expression.

Factorise: $5x^2 - 13x - 6$

Answer:

$(5x + 2)(x - 3)$

Working:

Use the product-sum test before writing the brackets.

$a = 5, \quad b = -13, \quad c = -6$

$a \times c = 5 \times -6 = -30$

$-15 \times 2 = -30 \quad \text{and} \quad -15 + 2 = -13$

Split the middle term using those two numbers.

$5x^2 - 13x - 6$

$= 5x^2 - 15x + 2x - 6$

Group the expression in pairs, then factor each group.

$= 5x(x - 3) + 2(x - 3)$

$= (5x + 2)(x - 3)$

Teacher answers & working

Factorising Quadratic Expressions · Coefficient of x squared

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E1

Factorise each expression.

Factorise: $15x^2 - 46x + 35$

Answer:

$(5x - 7)(3x - 5)$

Working:

Use the product-sum test before writing the brackets.

$a = 15, \quad b = -46, \quad c = 35$

$a \times c = 15 \times 35 = 525$

$-25 \times -21 = 525 \quad \text{and} \quad -25 + -21 = -46$

Split the middle term using those two numbers.

$15x^2 - 46x + 35$

$= 15x^2 - 25x - 21x + 35$

Group the expression in pairs, then factor each group.

$= 5x(3x - 5) - 7(3x - 5)$

$= (5x - 7)(3x - 5)$

E2

Factorise each expression.

Factorise: $30x^2 - 37x - 7$

Answer:

$(5x - 7)(6x + 1)$

Working:

Use the product-sum test before writing the brackets.

$a = 30, \quad b = -37, \quad c = -7$

$a \times c = 30 \times -7 = -210$

$5 \times -42 = -210 \quad \text{and} \quad 5 + -42 = -37$

Split the middle term using those two numbers.

$30x^2 - 37x - 7$

$= 30x^2 + 5x - 42x - 7$

Group the expression in pairs, then factor each group.

$= 5x(6x + 1) - 7(6x + 1)$

$= (5x - 7)(6x + 1)$

Teacher answers & working

Factorising Quadratic Expressions · Coefficient of x squared

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E3

Factorise each expression.

Factorise: $3x^2 - 19x + 28$

Answer:

$(3x - 7)(x - 4)$

Working:

Use the product-sum test before writing the brackets.

$a = 3, \quad b = -19, \quad c = 28$

$a \times c = 3 \times 28 = 84$

$-12 \times -7 = 84 \quad \text{and} \quad -12 + -7 = -19$

Split the middle term using those two numbers.

$3x^2 - 19x + 28$

$= 3x^2 - 12x - 7x + 28$

Group the expression in pairs, then factor each group.

$= 3x(x - 4) - 7(x - 4)$

$= (3x - 7)(x - 4)$